

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
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Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

M33 is a spiral galaxy in the local group of approximately $5 \cdot 10^{10} M_{\text{sun}}$ containing ~ 40 billion stars. It is the third largest galaxy in the local group and it is a major target for determining the Hubble constant, which defines the rate of expansion of the universe. To determine the rate of expansion, a distance to M33 is needed. Our program targets this distance measurement. We will take photometry of M33 cepheids with the Bok 90" Telescope utilizing the 90prime instrument in the g and i bands, with specific targeting of D33 J013401.2+303113.7 and D33 J013352.4+303844.2 on the nights of October 2, 2026, or October 3, 2026. These measurements will be combined with light curves of our targets from archival observations of a prior University of Arizona project by Max Moe. The Cepheid period-distance relation will then be used to determine M33's distance.

Cepheids are chosen specifically due to their use in the distance ladder. Cepheids are high mass, unstable, luminous stars. Repetitive expansion and contraction of their atmospheres result in an apparent brightness cycle observable from Earth-based telescopes. The period of the cycle is nearly contingent on the absolute brightness of the cepheid with a minor relation to metallicity. An empirical relation has thus been built from nearby cepheids to measure the distance to distant ones. By fitting the empirical relation to the light curves of multiple cepheids to obtain their absolute magnitudes, a distance to any observed cepheid can be obtained. This property is useful for deriving the distance to nearby galaxies due to the reliability of the relation and the possibility of fitting to multiple cepheids in different bands, yielding precise errors.

Our target cepheids have an apparent magnitude of ~ 19 . As such, any background produced by atmospherically scattered moonlight will significantly impact their visibility and thus their magnitude certainty. Only a dark or absent moon is appropriate for our program. Additionally, if photometry of our targets cannot be obtained due to poor weather, we will move forward with only the archival data.

To follow the work of the prior project, we will take 15 pointings of M33 in each band, ensuring the target cepheids appear in a minimum of 6 pointings. We request 3 minutes per pointing for a total of 90 minutes of observal time. The SNR we will gain from this is, based upon the observations of Marci et al. of M33 Cepheids, 360 for a 19-mag cepheid.

Our additional data points will provide additional fitting of the light curves, improving the distance measurement from the prior project. This will improve mapping of the local group and provide a more accurate distance for measurement of the Hubble constant.

Target List

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
M33	01:33:50.90	+30° 39' 36.63"	5.72	Galaxy	Sept-Jan

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D33 J013401.2+303113.7	01:34:01.20	+30° 31' 13.40"	19.096	Cepheid	Sept-Jan
D33 J013352.4+303844.2	01:33:52.45	+30° 38' 44.08"	19.380	Cepheid	Sept-Jan

References

L. M. Macri *et al* 2001 *AJ* **121** 861